TSM 099 Arithmetic Test

Practice 2 | Version 202610122020.17

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Evaluate at (a) x = 3 and (b) x = -3. Show substitution. $(x+3)(2x-2)$	1. (a) 10 pts (b) 10 pts
2. Simplify the square root. $\sqrt{75}$	2. Radical 20 pts
3. Evaluate each side separately at x = 7, -7, 0. Record the left value, right value, and whether they are equal. State what the tested values establish; do not solve the equation. $\frac{x^2-9}{x-3} = \frac{2x^2+11x+12}{x+4}$	3. x = 7: left / right / equal? 4 pts x = -7: left / right / equal? 4 pts x = 0: left / right / equal? 4 pts Conclusion 8 pts
4. Evaluate the principal square root in each part. $(a)\quad \sqrt{-64}$ $(b)\quad \sqrt{36}$	4. (a) 10 pts (b) 10 pts
5. Reduce each fraction to lowest terms. $(a)\quad \frac{60}{84}$ $(b)\quad \frac{25}{35}$	5. (a) 10 pts (b) 10 pts

Instructor answer key: TSM 099 Arithmetic Test

Practice 2 | Version 202610122020.17

Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

24

10 pts

(b)

0

10 pts

2. Radical

$5\sqrt{3}$

20 pts

3. x = 7: left / right / equal?

Left 10; right 17; no

4 pts

x = -7: left / right / equal?

Left -4; right -11; no

4 pts

x = 0: left / right / equal?

Left 3; right 3; yes

4 pts

Conclusion

Only the listed matching value works; the equation is not an identity.

8 pts

4. (a)

8i

10 pts

(b)

6

10 pts

5. (a)

$\frac{5}{7}$

10 pts

(b)

$\frac{5}{7}$

10 pts